



Alignment in
Armenian

R. Meyer
(Oxford)

bit.ly/ArmAlign18

Prolegomena

Armenia(n)

Alignment

Arm. PF

WMLr. PST

Model → Target

Stats

Conclusions

Alignment change and changing alignments: the Armenian perfect and its Iranian model

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Variation and Change in the ancient Indo-European Languages
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Outline

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- 1 Prolegomena
 - Armenia(n): what, where, when, why?
 - Morphosyntactic alignment patterns
- 2 The Armenian *-eal* perfect
- 3 The West Middle Iranian past tense
- 4 Pivot matching & pattern replication
- 5 Statistics: diachronic developments in the Armenian perfect
- 6 Conclusions



Armenia(n) in brief I

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- ▶ Armenian is an Indo-European language, maybe most closely related to Greek, Phrygian, Albanian (Pontic Group; for isoglosses, cf. HOLST 2009; more critically CLACKSON 1994)
- ▶ originally spoken in the Armenian Highland, delimited by the Caucasus and Taurus Mountains, the Upper Euphrates and Media Atropatene
- ▶ recognised first as a people in the Behistun inscription (c. 520 BCE)
- ▶ first attested c. 480 CE (Sinai inscriptions), first MS c. 9th century



Armenia(n) in brief II

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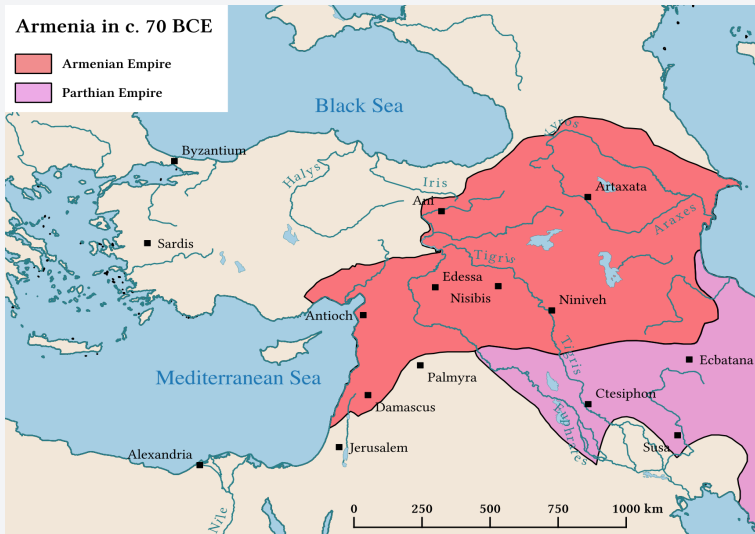
Model → Target

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Conclusions

Armenia in c. 70 BCE

- Armenian Empire
- Parthian Empire





Armenia(n) in brief III

Alignment in
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Conclusions

Armenia in c. 379 CE

- Kingdom of Armenia
- Roman Province Armenia Minor
- Sasanian Empire





Armenian & Iranian

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- ▶ thought to belong to Iranian phylum until HÜBSCHMANN (1875)
- ▶ strong influence from Iranian languages (Old and Western Middle Iranian) in many categories: phonology, lexicon, derivational morphology, syntax
- ▶ since MEILLET (1911–2) clear that main influence from Parthian (North West Middle Iranian)
- ▶ statistics suggest that only 22% of Classical Armenian vocabulary (as per HÜBSCHMANN's 1897 analysis) are *echtarmenisch*, more than 30% Parthian (BELARDI 2003:98–102)



- ▶ 6th – 3rd century BCE: Achaemenid satrapy under Iranian rule, then independent (Orontid dynasty)
- ▶ mid-3rd century BCE – early 1st century CE: independent kingdoms, unified in an Empire in 1st century BCE (Artaxiads)
- ▶ 66 – 428 CE: Arsacid rule (junior branch of Parthian ruling dynasty)
- ▶ intermittently Roman protectorate or occupied by Sasanian forces (at least in part)

thus ruled by Iranians for c. 1,000 years

- for further reference, cf. GARSOÏAN (1997*a,b,c*); RUSSELL (1997)



Armenians & Iranians II

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Conclusions

- ▶ establishment of a *hereditary* dynasty from early 3rd century CE
- ▶ fall of the Parthian Empire (224 CE), ensuing political quarrels with the Sasanian successor state
- ▶ Christinisation of Armenia in c. 301–14 – as opposed to Sasanian Zoroastrianism
- ▶ frequent Armenian–Iranian intermarriage, tutelage system
- ▶ ‘death’ of Parthian in the region

thus Parthian minority superstrate shifts to Armenian

- ▶ arguably not the first shift in Parthian history (HENNING 1958; LECOQ 1986)
- ▶ cp. situation in post-conquest Britain (BERNDT 1965; SCHENDL 2000; SHORT 1980)



Iranian influence: lexicon I

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Conclusions

- ▶ Pth. loans are pervasive throughout lexicon (specialised and basic), incl. closed classes

Pth. loan words in Arm.

	Arm.	Pth.
‘army’	<i>zawr</i>	<i>z’wr</i> /zāwar/
‘root’	<i>bun</i>	<i>bwn</i> /bun/
‘white’	<i>spitak</i>	<i>‘spyd</i> /ispēd/
‘1,000’	<i>hazar</i>	<i>hz’r</i> /hazār/
‘because of’	<i>vasn</i>	<i>wsn’d</i> /wasnāδ/

- ▶ Middle Persian and Old Iranian loans restricted to specialised vocabulary (esp. administration, military)
- ▶ Pth. phonemes are imported with lexical items (spreading into native words)
- ▶ cf. BOLOGNESI (1960); MEYER (2017); SCHMITT (1983)



- ▶ calques of various types occur, esp. in complex predicates and nominal compounds

Pth. calques in Arm.

	Arm.	Pth.	lit.
‘friend’	<i>barekam</i>	<i>šyrg’mg</i> /šīrgāmag/	‘well-wisher’
‘prisoner’	<i>jerbakal</i>	<i>dstgrb</i> /dastgraß/	‘taken by the hand’
‘to remove’	<i>heri ainel</i>	<i>dwr kr-</i> /dūr kar-/	‘to make far’

- ▶ complex predicates frequently only attested in Modern Persian evidence owing to limited Middle Iranian data
- ▶ further reference: BOLOGNESI (1961, 2006)



Iranian influence: syntax

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- ▶ usage of the intensifier Arm. *inkʻn* ‘self’ also as anaphora and (to a lesser extent) as switch-function marker – exact parallel in Pth. and MP (MEYER 2013)
- ▶ canonical reflexive Arm. *anjn iwr* modelled on Pth. *xwys̥ gryw* /*xwēš grīw*/ ‘own body/self’
- ▶ use of complementiser Arm. *(e)tʻē* also as quotative and indirect statement/question marker, just like WMIr. *kw* /*kū*/ (rare elsewhere; MEYER 2017)



Morphosyntactic alignment types (selection)

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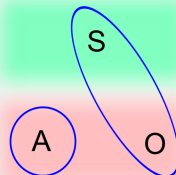
Conclusions

*intransitive
clause*

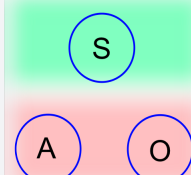
*transitive
clause*



nominative-
accusative



ergative-
absolutive



tripartite



Alignment in Armenian I

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Stats

Conclusions

- ▶ for the most part, Armenian has NOM-ACC alignment, verbal agreement with NOM

- ▶ construction: (OBJ-marker +) ACC direct object + NOM subject/agent + verb

(1) ... *elanēr na i telis mehenac'n* ...
go.3.SG.PST 3.SG.NOM (in)to place.ACC.PL temple.GEN.PL

‘[And after this] he went to the sites of the temples ...’ (Ag. §814)

(2) *du es ayn, or kotorec'er*
2.SG.NOM be.2.SG.PRS DEM.NOM.SG REL.NOM.SG destroy.3.SG.AOR
z=Aris aysč'ap' ams ...
OBJ=Aryan.ACC.PL so-many year.ACC.PL

‘It is you, who has destroyed the Aryans for so many years ...’
(P'B IV.54)

- ▶ The perfect intransitive and passive follow this pattern, too (for the most part)



Alignment in Armenian II

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Prolegomena

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Model → Target

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Conclusions

but perfect transitive construction:

(**OBJ-marker** +) **ACC direct object** + **GEN agent** + **PTCP** (+ **3.SG copula**)

- (3) *ew gteal Yisusi ēš mi*
 and **find.PTCP** **PN.GEN.SG** **donkey.NOM-ACC.SG** **INDEF**

‘And Jesus found a donkey’ (Jn. 12:14)

- (4) ... *zi ēr paheal z=mez amenazawr*
 ... for **3.SG.PST** **preserve.PTCP** **OBJ=1.PL.ACC** **almighty**
ajoyr
right.GEN.SG

‘... for his almighty right [hand] has preserved us’
 (Agat‘angelos §186)



Alignment in Armenian III

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Prolegomena

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Model → Target

Stats

Conclusions

- (1) Armenian has split-tripartite alignment: NOM-ACC in non-perfective, NOM-GEN-ACC in perfective aspect
 - (2) copula agreement in the perfect, where present, is with Ø
 - (3) on occasion, GEN subjects and NOM agents can be found in the perfect – contrary to expectations
-
- ▶ previous, non-contact explanations struggle to explain co-existence of transitive and intransitive constructions, form of copula (cf. e.g. BENVENISTE 1952; KÖLLIGAN 2013; MEILLET 1936; STEMPEL 1983; WEITENBERG 1986)



Alignment in Armenian

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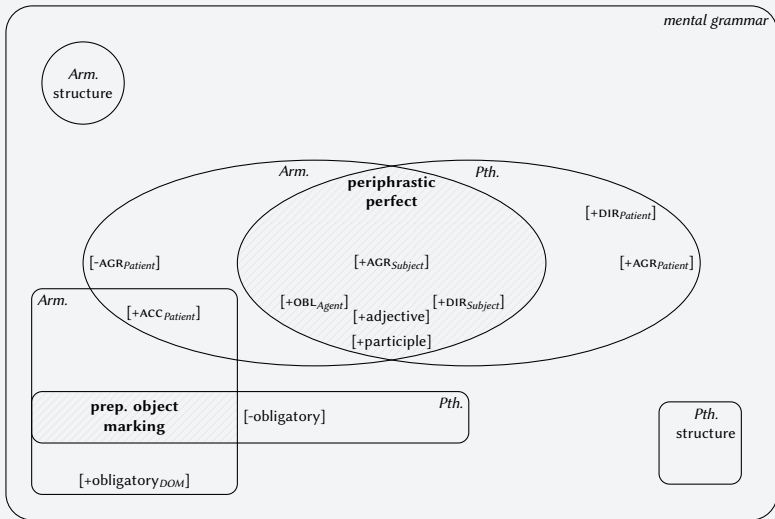
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Model $\rightarrow$ Target





Statistics

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Model → Target

Stats

Conclusions

► given the above, at least two things might be expected:

- (1) the increase, under system pressure, of finite verb usage (copula);
 - (2) the move from tripartite to NOM-ACC alignment, like from ERG-ABS to tripartite before;
- the latter is evident since from the 8th century CE onwards, NOM-ACC is the standard alignment pattern even in the perfect



Stat I: Incidence of non-standard argument marking

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Alignment

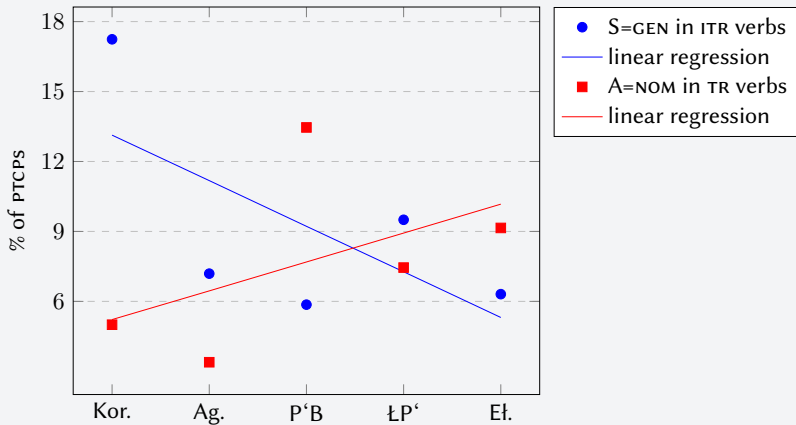
Arm. PF

WMLr. PST

Model $\rightarrow$ Target

Stats

Conclusions





Stat II: Incidence and trend of the copula

Alignment in
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Alignment

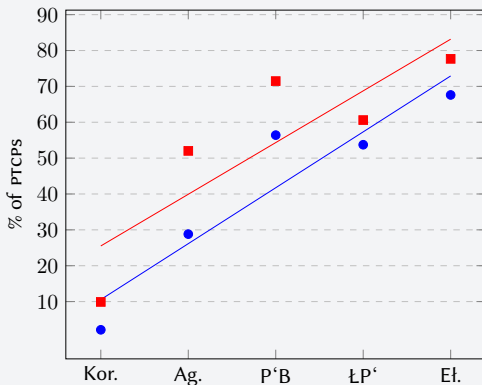
Arm. PF

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Stats

Conclusions



- % TR.ACT perfects with copula
- linear regression
- % of all perfects with copula
- linear regression



Conclusions

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Stats

Conclusions

- ▶ language contact with Parthian provides a good basis for explaining the alignment change in Classical (and pre-literary) Armenian
- ▶ pivot matching and pattern replication, esp. under the socio-historical circumstances listed above, better explain the syntax of the perfect than internal approaches alone
- ▶ statistics corroborate this alignment change, though the corpus needs to be expanded for a fuller picture
- ▶ alignment changes of this type are well evidenced in the region (cf. e.g. CIANCAGLINI 2008; KHAN 2007)
- ▶ further research is required to expand our understanding of Parthian-based pattern replication in Classical Armenian



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Conclusions

Thank you for your attention!



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BELARDI, W. (2003) *Elementi di armeno aureo*, volume I, Rome: Il Calamo.

BENVENISTE, É. (1952) “La construction passive du parfait transitif,” *Bulletin de la Société Linguistique de Paris* 48, 52–62.

BERNDT, R. (1965) “The linguistic situation in England from the Norman conquest to the loss of Normandy,” *Philologica Pragensia* 8, 145–163.

BOLOGNESI, G. (1960) *Le fonti dialettali degli imprestiti iranici in armeno*, Milan: Società Editrice Vita e Pensiero.

—— (1961) “Sul valore e l’origine di alcuni suffissi armeni,” *Atti del Sodalizio Glottologico Milanese* 13–14, 16–17.

—— (2006) “L’influsso iranico, greco e latino sul lessico armeno,” in R. GUSMANI and R. BOMBI (eds.), *Studi linguistici in onore di Roberto Gusmani*, Alessandria: Edizioni dell’Orso, 263–274.

CIANCAGLINI, C.A. (2008) *Iranian Loanwords in Syriac*, Wiesbaden: Dr. Ludwig Reichert.



References II

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CLACKSON, J. (1994) *The Linguistic Relationship between Armenian and Greek*, Oxford/Cambridge, MA: Blackwell.

GARSOÏAN, N.G. (1997a) “The Aršakuni Dynasty (A.D. 12–[180?]-428),” in R.G. HOVANNISIAN (ed.), *The Armenian People from Ancient to Modern Times, vol. 1: The Dynastic Periods, from Antiquity to the Fourteenth Century*, New York: St Martin’s Press, 63–94.

—— (1997b) “The Emergence of Armenia,” in R.G. HOVANNISIAN (ed.), *The Armenian People from Ancient to Modern Times, vol. 1: The Dynastic Periods, from Antiquity to the Fourteenth Century*, New York: St Martin’s Press, 37–62.

—— (1997c) “The Marzpanate,” in R.G. HOVANNISIAN (ed.), *The Armenian People from Ancient to Modern Times, vol. 1: The Dynastic Periods, from Antiquity to the Fourteenth Century*, New York: St Martin’s Press, 95–116.

HENNING, W.B. (1958) “Mitteliranisch,” in *Handbuch der Orientalistik*, volume I/IV.1, Brill, 20–129.



References III

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- HOLST, J.H. (2009) *Armenische Studien*, Wiesbaden: Harrassowitz.
- HÜBSCHMANN, H. (1875) “Ueber die stellung des armenischen im kreise der indogermanischen sprachen,” *Zeitschrift für vergleichende Sprachforschung auf dem Gebiete der Indogermanischen Sprachen* 23 (1), 5–49.
- (1897) *Armenische Grammatik. 1. Theil: Armenische Etymologie*, Leipzig: Breitkopf & Härtel.
- KHAN, G. (2007) “Grammatical borrowing in North-eastern Neo-Aramaic,” in Y. MATRAS and J. SAKEL (eds.), *Grammatical Borrowing in Cross-Linguistic Perspective*, Berlin: Mouton de Gruyter, 197–214.
- KÖLLIGAN, D. (2013) “Non-canonical subject marking. Genitive subjects in Classical Armenian,” in I.A. SERŽANT and L.I. KULIKOV (eds.), *The Diachronic Typology of Non-Canonical Subjects*, Amsterdam: John Benjamins, 73–90.



References IV

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LECOQ, P. (1986) “Aparna,” in E. YARSHATER (ed.), *Encyclopædia Iranica*, volume II/2, London: Routledge & Kegan Paul, 151.

MATRAS, Y. (2009) *Language Contact*, Cambridge: Cambridge University Press.

MATRAS, Y. and SAKEL, J. (2007) “Investigating the mechanisms of pattern replication in language convergence,” *Studies in Language* 31 (4), 829–865.

MEILLET, A. (1911–2) “Sur les mots iraniens empruntés par l’arménien,” *Mémoires de la Société Linguistique de Paris* 17, 252–250.

——— (1936) *Esquisse d’une grammaire comparée de l’arménien classique*, second entirely revised edition, Vienna: Imprimerie des PP. Mékhitaristes.

MEYER, R. (2013) “Armeno-Iranian Structural Interaction: The Case of Parthian wxd, Armenian ink‘n,” *Iran and the Caucasus* 17 (4), 401–425.



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——— (2017) *Iranian-Armenian language contact in and before the 5th century CE. An investigation into pattern replication and societal multilingualism*, Ph.D. thesis, University of Oxford, Oxford.

——— (2019) “The Relevance of Typology for Pattern Replication,” *Journal of Language Contact* 12 (1).

RUSSELL, J. (1997) “The Formation of the Armenian Nation,” in R.G. HOVANNISIAN (ed.), *The Armenian People from Ancient to Modern Times, vol. 1: The Dynastic Periods, from Antiquity to the Fourteenth Century*, St Martin’s Press, 19–36.

SCHENDL, H. (2000) “Linguistic Aspects of Code-Switching in Medieval English Texts,” in D.A. TROTTER (ed.), *Multilingualism in Later Medieval Britain*, Rochester, NY: D. S. Brewer, 77–92.

SCHMITT, R. (1983) “Iranisches Lehngut im Armenischen,” *Revue des Études Arméniennes* 17, 73–112.

SHORT, I. (1980) “On Bilingualism in Anglo-Norman England,” *Romance Philology* 33 (4), 467–479.



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STEMPEL, R. (1983) *Die infiniten Verbalformen des Armenischen*, Frankfurt a.M./Bern/New York: Peter Lang.

THOMASON, S.G. (2003) “Contact as a Source of Language Change,” in B.D. JOSEPH and R.D. JANDA (eds.), *Handbook of Historical Linguistics*, Malden, MA: Blackwell, 687–712.

WEITENBERG, J. (1986) “Infinitive and Participle in Armenian,” *Annual of Armenian Linguistics* 7, 1–26.